No. of Printed Pages : 4 220843

Roll No. ....

4th Sem / Computer, Computer (For Speech and Hearing Impaired)

Subject : Data Structures Using C

Time : 3 Hrs. M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 Which variables are accessed by all modules of the C program.

a) Local Variables b) Global variables

c) Both a and b d) None of the above

Q.2 The nodes of the tree having same parent are called

a) Non terminal nodes b) Parent nodes

c) Siblings d) Terminal nodes

Q.3 Which of the following is/are Non-contiguous data structures

a) Stack b) Array

c) Linked List d) Trees

Q.4 Which of the following sorting algorithms falls in the category of Divide and Conquer technique

a) Selection sort b) Quick sort

c) Bubble sort d) All of these

Q.5 Which of the following properties is followed by a Queue.

a) FIFO b) LIFO

c) FILO d) Any of these

Q.6 Which of the following data structures is used to evaluate a expression given In postfix notation?

a) Queue b) Tree

c) Stack d) Linked list

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 The variable which stores the address of another variable is called______

Q.8 Define Data Structures.

Q.9 Define the node structure of a singly linked.

Q.10 Give the formula for incrementing the FRONT in circular queue.

Q.11 A node in a binary tree can have any numbers of children.(True/False)

Q.12 Name the two methods of searching for presence of element in the given list.

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Give any four differences between an Array and a Linked List

Q.14 Describe various types of data in C language.

Q.15 Explain contiguous and non-contiguous data structures

Q.16 Give algorithm for adding a node at the beginning of the doubly linked list

Q.17 Give the differences between linear and binary searching

Q.18 Give algorithm for deleting an element from a stack?

Q.19 What is the limitation of a linear queue? How is it removed?

Q.20 Create the tree for the following expression and traverse it in preorder form

A*B/C* D+ E/ F+G -H

Q.21 Sort the following list of element using HEAP Sort.

14 26 22 19 11 53 4 9

Q.22 Write short notes on any one of these

a) Recursion

b) Tap down approach

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Define Array. Give the different types of Arrays? Explain how element of two dimensional arrays are stored in memory?

Q.24 Write algorithm for bubble sort? Explain with a suitable example.

Q.25 Explain the three ways of traversing in a binary tree with algorithm of each?